Microeconomics I

Class 9 — Practice for the Part I exam

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Practicalities

- Problem set 2: returned in this class, graded.
- Today: three problems by hand, plus an optional fourth.
  In pairs first, then the board. The problem list is on the course page; worked solutions there after class.
- Class 9's theory is **not** in the Part I exam.
  Arrow–Pratt, comparing risk aversion, insurance, stochastic dominance: optional reading on the course page after class; material for the final exam on 18 December.
- **Part I exam: Wednesday 28 October, 09:30–11:20, Room 1030.**
  Photo ID and a pen. A simple calculator is allowed but never needed. Phones and other electronics switched off and put away. Any handwritten notes, also as photocopies or printouts; no typed notes or other printed material.

Problem 1 · Cobb–Douglas, demand known: v, e, h, CV and EV

$u(x_1, x_2) = x_1\, x_2^2$

(a) Walrasian demand $x(p, w)$ by the Cobb–Douglas share rule; its value at $p = (1, 1)$, $w = 6$.

(b) The indirect utility function $v(p, w)$.

(c) The expenditure function $e(p, u)$, by inverting v.

(d) The Hicksian demand $h(p, u)$ by Shephard's lemma; check that $h(p, u) = x\big(p, e(p, u)\big)$ for all p, u.

(e) Good 2 dearer: $p^0 = (1, 1)$ to $p^1 = (1, 8)$; $w = 6$. The new bundle; the substitution effect $h_2(p^1, u^0) - x_2(p^0, w)$ and the wealth effect $x_2(p^1, w) - h_2(p^1, u^0)$; $h_1(p^1, u^0)$ against $x_1(p^0, w)$; $\mathrm{CV} = w - e(p^1, u^0)$, $\mathrm{EV} = e(p^0, u^1) - w$.

In pairs: part (a). Then the board.

Problem 1 · Cobb–Douglas, demand known: v, e, h, CV and EV

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In pairs: part (a). Then the board.

Results

(a) $x(p, w) = \left(\frac{w}{3p_1}, \frac{2w}{3p_2}\right) = (2, 4)$

(b) $v(p, w) = \frac{4w^3}{27\, p_1 p_2^2}$ $(= 32$ here$)$

(c) $e(p, u) = 3\left(p_1 p_2^2\, u/4\right)^{1/3}$

(d) $h_1 = \frac{e(p,u)}{3p_1}$, $h_2 = \frac{2\, e(p,u)}{3p_2}$

(e) new bundle $(2, \frac{1}{2})$; $h(p^1, u^0) = (8, 2)$
on x_2: substitution -2, wealth $-\frac{3}{2}$
(total $-\frac{7}{2}$); h_1: $2 \to 8$
$\mathrm{CV} = -18$, $\mathrm{EV} = -\frac{9}{2}$

Problem 2 · $u = 2\sqrt{x_1} + 3\sqrt{x_2}$: solve it by hand

$u(x_1, x_2) = 2\sqrt{x_1} + 3\sqrt{x_2}$

(a) The Lagrangian of: maximize $u(x)$ subject to $p_1 x_1 + p_2 x_2 \leq w$ (one multiplier); the first-order conditions; solve them for $x(p, w)$. Why is the optimum interior? Why are the first-order conditions sufficient? Its value at $p = (1, 3)$, $w = 56$.

(b) $v(p, w)$, as a single square root.

(c) $e(p, u)$.

(d) $h(p, u)$ by Shephard's lemma.

(e) Prices move from $p^0 = (1, 3)$ to $p^1 = (8, 3)$; $w = 56$. The new bundle and utility; CV and EV; the substitution and wealth effects on x_1, as in Problem 1.

In pairs: set up part (a). Then the board: (a) and (b) now, (c) to (e) after the break.

Problem 2 · $u = 2\sqrt{x_1} + 3\sqrt{x_2}$: solve it by hand

$u(x_1, x_2) = 2\sqrt{x_1} + 3\sqrt{x_2}$

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Results

(a) $\frac{1}{\sqrt{x_1}} = \lambda p_1$, $\frac{3}{2\sqrt{x_2}} = \lambda p_2$: $\sqrt{x_2/x_1} = \frac{3p_1}{2p_2}$

$x_1 = \frac{4w\,p_2}{p_1(4p_2+9p_1)}$, $x_2 = \frac{9w\,p_1}{p_2(4p_2+9p_1)}$; (32, 8) here

interior: $\partial u/\partial x_i \to \infty$ as $x_i \to 0$

sufficient: u is concave, the constraint linear

(b) $v(p, w) = \sqrt{\frac{w\,(4p_2+9p_1)}{p_1\,p_2}}$ $(= 14\sqrt{2}$ here)

Break

Back at 11:00

Problem 2, continued · $u = 2\sqrt{x_1} + 3\sqrt{x_2}$

$u(x_1, x_2) = 2\sqrt{x_1} + 3\sqrt{x_2}$

(a) $x_1 = 4wp_2/\big(p_1(4p_2 + 9p_1)\big)$, $x_2 = 9wp_1/\big(p_2(4p_2 + 9p_1)\big)$; $(32, 8)$ at $p = (1, 3)$, $w = 56$.

(b) $v(p, w) = \sqrt{w\,(4p_2 + 9p_1)/(p_1\,p_2)}$.

(c) The expenditure function $e(p, u)$.

(d) The Hicksian demand $h(p, u)$ by Shephard's lemma.

(e) Prices move from $p^0 = (1, 3)$ to $p^1 = (8, 3)$; $w = 56$. The new bundle and utility (tip: work with $(u^0)^2$); $\mathrm{CV} = w - e(p^1, u^0)$ and $\mathrm{EV} = e(p^0, u^1) - w$; the substitution effect $h_1(p^1, u^0) - x_1(p^0, w)$ and the wealth effect $x_1(p^1, w) - h_1(p^1, u^0)$.

In pairs: part (c). Then the board.

Problem 2, continued · $u = 2\sqrt{x_1} + 3\sqrt{x_2}$

$u(x_1, x_2) = 2\sqrt{x_1} + 3\sqrt{x_2}$

(a) $x_1 = 4wp_2/\big(p_1(4p_2 + 9p_1)\big)$, $x_2 = 9wp_1/\big(p_2(4p_2 + 9p_1)\big)$; $(32, 8)$ at $p = (1, 3)$, $w = 56$.

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In pairs: part (c). Then the board.

Results

(c) $e(p, u) = \frac{u^2\, p_1\, p_2}{4p_2 + 9p_1}$

(d) $h_1 = \frac{4u^2\, p_2^2}{(4p_2 + 9p_1)^2}$, $h_2 = \frac{9u^2\, p_1^2}{(4p_2 + 9p_1)^2}$

(e) new bundle $(1, 16)$, $u^1 = 14$; $(u^0)^2 = 392$
$\mathrm{CV} = -56$, $\mathrm{EV} = -28$
$h(p^1, u^0) = (2, 32)$
substitution -30, wealth -1 (total -31)

Problem 3 · Expected utility: a two-stage game

Martina maximizes expected utility with $u(x) = \sqrt{x}$ ($x \geq 0$ euros). She throws a fair die. On 1 or 2: lottery A, 144 euros with probability $\frac{1}{4}$, else 0. On 3, 4, 5 or 6: lottery B, 81 or 9 euros, $\frac{1}{2}$ each.

(a) The reduced lottery: each prize with its probability.

(b) $E[x]$, $E[u(x)]$, her certainty equivalent c, and $\mathrm{RP} = E[x] - c$. Her attitude to risk: where does it show?

(c) Offer: s euros for sure in place of B; A stays. Why, by the independence axiom, does A not matter? For which s does she weakly prefer the offer?

(d) Pau is risk neutral: for which s? A strict risk lover: a larger or a smaller s than Pau? Why?

In pairs: part (a). Then the board.

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In pairs: part (a). Then the board.

Results

(a) 0 euros: $\frac{1}{4}$, 9: $\frac{1}{3}$, 81: $\frac{1}{3}$, 144: $\frac{1}{12}$

(b) $E[x] = 42$ euros, $E[u] = 5$ utils,
$c = 25$ euros, $\mathrm{RP} = 17$ euros;
u concave: risk averse, so $c < E[x]$

(c) s against B only: $s \geq c_B = 36$ euros

(d) Pau: $s \geq E_B[x] = 45$ euros;
a strict risk lover needs more than 45 euros

Problem 4 (optional) · Properties of preferences

Joan values good 1 on its own, and good 2 only together with good 1, one unit with one unit:

$$u(x_1, x_2) = x_1 + \min\{x_1, x_2\} \quad \text{on } \mathbb{R}^2_+.$$

Are her preferences **(a)** monotone? **(b)** strongly monotone? **(c)** locally non-satiated (LNS)? **(d)** convex?

Each: *yes*, with a reason, or *no*, with a counterexample.

Monotone: $y \gg x$ implies $y \succ x$. Strongly monotone: $y \geq x$ and $y \neq x$ imply $y \succ x$ (MWG Def. 3.B.2).

LNS: arbitrarily close to every x there is a $y \succ x$ (Def. 3.B.3). Convex: every set $\{y : y \succsim x\}$ is convex (Def. 3.B.4).

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Results

(a) Yes: $y \gg x$ raises x_1 and $\min\{x_1, x_2\}$

(b) No: $(1,3) \geq (1,2)$, yet $u(1,3) = u(1,2) = 2$

(c) Yes: monotone preferences are LNS

(d) Yes: $u = \min\{2x_1, x_1 + x_2\}$ is concave, so every upper contour set is convex

Good luck on Wednesday 28 October

Today by hand: demand $\to v \to e \to h \to$ CV and EV; a compound lottery, its certainty equivalent and risk premium.

Part I exam: Wednesday 28 October, Room 1030.

It runs from 09:30 to 11:20.

Bring photo ID and a pen. A simple calculator is allowed but never needed. Phones switched off and put away. Any handwritten notes, also as photocopies or printouts; no typed notes or other printed material.

Revise: Sextant, pkocourek.com/sextant/?c=micro-1

Worked solutions and the optional reading (class 9's theory): pkocourek.com/teaching/micro-1, after class.

Part II (theory of the firm, general equilibrium) is taught by Olga Chiappinelli from November.

Thank you, and good luck.